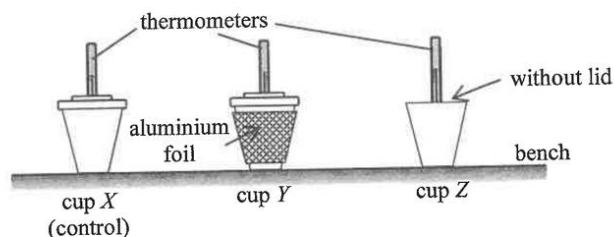


Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

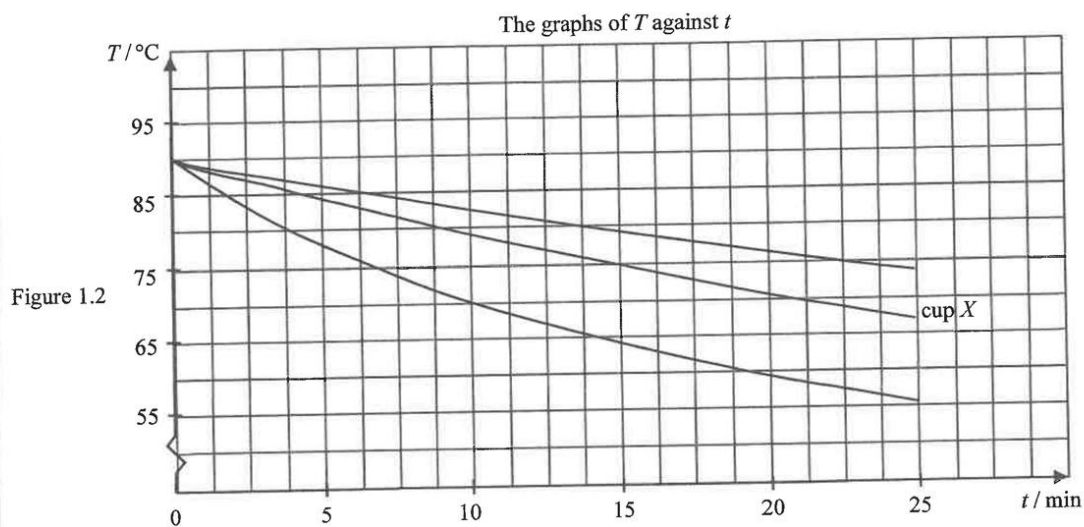
1. A student conducts an experiment to study how to best retain the warmth of water by using identical paper cups *X*, *Y* and *Z* shown in Figure 1.1. Each cup is filled with 250 cm³ of hot water and cup *X* serves as a control.

Figure 1.1



cup	wrapping	with a lid
<i>X</i>	no	yes
<i>Y</i>	aluminium wrapping	yes
<i>Z</i>	no	no

When the water temperature is 90 °C, the student starts taking thermometer readings every minute. Figure 1.2 shows the variation of temperature (*T*) of water in the cups with time (*t*) elapsed.



- (a) Suggest a reason why this experiment begins with the **same initial water temperature** (90 °C). (1 mark)

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Answers written in the margins will not be marked.

Book 1 section 1.4: DSE long questions

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